

ENGINEERING

# Deployment Standard Operating Procedure

CI/CD pipeline, environments, release windows, canary rollout, rollback and incident response

|                |                                                       |                |                         |
|----------------|-------------------------------------------------------|----------------|-------------------------|
| Document ID    | NX-ENG-002                                            | Version        | 3.1                     |
| Document owner | Head of DevOps and SRE                                | Effective date | 1 April 2026            |
| Applies to     | All engineers, QA, release managers and on-call staff | Classification | Internal - Confidential |

**Abstract.** This SOP defines how software is built, tested and released to production at Nexora Technologies. It describes the environments, the automated CI/CD pipeline, approval gates, release windows, the canary deployment strategy, criteria for automatic rollback, incident severity levels and the on-call rotation. Following this procedure keeps production changes traceable and reversible.

**Keywords:** CI/CD, deployment, canary release, rollback, incident severity, on-call, change management

## 1. Scope and Principles

- Every production change is traceable to an approved pull request and a change record (CHG ticket) in Jira.
- Deployments are automated; engineers do not deploy from laptops or edit production resources manually.
- Every release must be reversible within **15 minutes**.
- Infrastructure is defined as code (Terraform) and applications are deployed with Argo CD on Amazon EKS.

## 2. Environments

Table 1. Nexora environments.

| Environment    | Purpose                                                        | Deploy trigger                             | Approval                          |
|----------------|----------------------------------------------------------------|--------------------------------------------|-----------------------------------|
| Development    | Feature testing by developers; synthetic data                  | Every merge to a feature branch (optional) | None                              |
| Staging        | Integration, regression and performance tests; anonymised data | Automatic on merge to main                 | None                              |
| Pre-production | Final validation with production-like configuration            | Manual trigger by release manager          | QA lead                           |
| Production     | Live customer traffic                                          | Manual approval after staging tests pass   | Release manager and service owner |

## 3. CI/CD Pipeline

The pipeline runs in GitHub Actions. A change moves through the stages in Figure 1; any failing stage stops the pipeline and notifies the author. Code must meet the review and coverage rules in the Coding Standards (NX-ENG-003) before it can be merged.

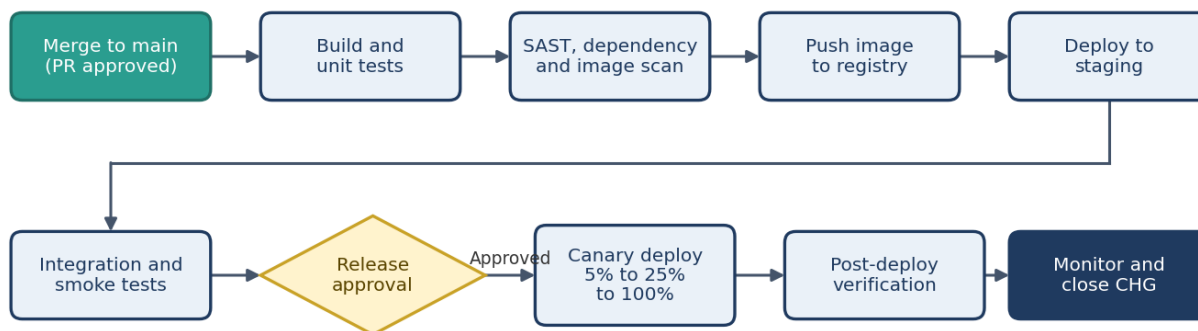


Figure 1. Nexora CI/CD pipeline from merge to production monitoring.

## 4. Release Windows and Approvals

Table 2. Release windows (Pakistan Standard Time).

| Release type               | Allowed window                                                           | Approval required                                                         |
|----------------------------|--------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Standard release           | Tuesday and Thursday, 10:00 AM to 2:00 PM                                | Release manager and service owner                                         |
| Minor configuration change | Monday to Thursday, 10:00 AM to 4:00 PM                                  | Service owner                                                             |
| Emergency hotfix           | Any time, including nights and weekends                                  | On-call lead and CTO (or delegate);<br>CHG ticket created within 24 hours |
| Blackout                   | Friday after 12:00 PM, weekends, public holidays,<br>and the week of Eid | Standard releases are not allowed                                         |

## 5. Deployment Strategy

Stateless services use a **canary rollout**: the new version first receives 5% of traffic for 15 minutes, then 25% for 15 minutes, and finally 100% (Figure 2). The API Gateway uses blue-green deployment. Database migrations must be backward compatible with the previous application version (expand, migrate, contract) so that a rollback never requires a schema rollback.

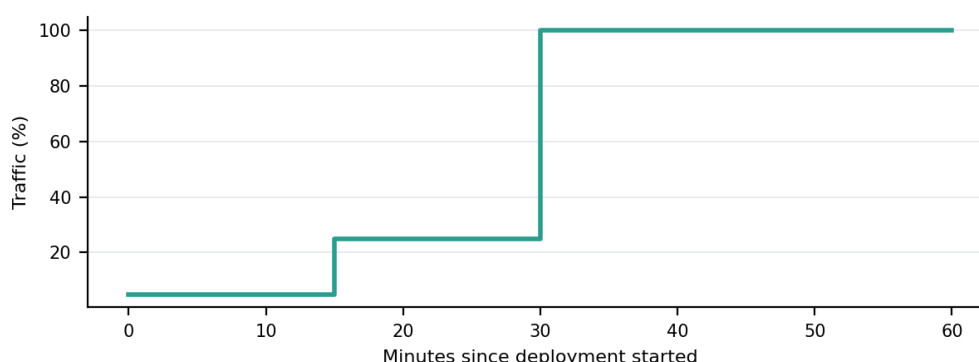


Figure 2. Canary rollout schedule: 5% for 15 minutes, 25% for 15 minutes, then full rollout.

## 6. Rollback

Automated monitoring compares the canary against the baseline. A rollback is triggered automatically, or by the release manager, if **any** of the following occurs:

- HTTP 5xx error rate above **2%** for 5 consecutive minutes.
- p95 latency above **2 times** the baseline for 5 consecutive minutes.
- Failed health checks on more than 20% of new pods, or a crash loop.

- A SEV1 or SEV2 defect that is confirmed to originate from the release.

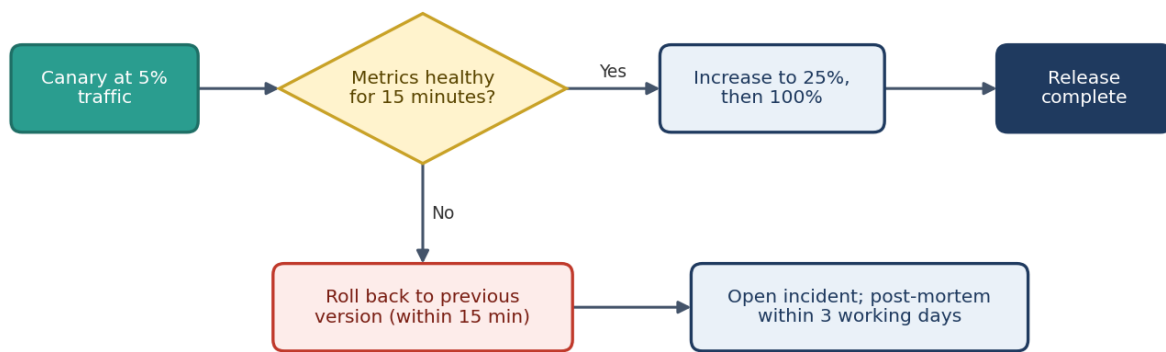


Figure 3. Rollback decision flow during a canary release.

## 7. Incident Severity and On-Call

Table 3. Incident severity levels.

| Level | Definition                                            | Acknowledge within | Update frequency |
|-------|-------------------------------------------------------|--------------------|------------------|
| SEV1  | Complete outage or data loss affecting many customers | 15 minutes         | Every 30 minutes |
| SEV2  | Major feature unavailable or severe degradation       | 30 minutes         | Every 60 minutes |
| SEV3  | Minor feature impaired; workaround exists             | 4 hours            | Daily            |
| SEV4  | Cosmetic issue or low-impact defect                   | Next business day  | As needed        |

The on-call rotation is weekly and handled by senior engineers (G3 and above) and a secondary engineer in G2. The primary on-call must be reachable within 10 minutes and able to reach a laptop within 30 minutes. On-call allowances are described in the Benefits Guide (NX-HR-003). SEV1 and SEV2 incidents require a written post-mortem within 3 working days, following a blameless format.

## 8. Checklists

### 8.1 Pre-Deployment

- Change ticket approved and linked to the pull request; release notes prepared.
- All pipeline stages passed on the exact image being released.
- Rollback plan and database migration compatibility confirmed.
- Customer-facing teams notified for changes visible to customers.
- Release is inside the permitted window in Table 2.

### 8.2 Post-Deployment

- Smoke tests pass and dashboards show error rate and latency within normal range for 30 minutes.
- Feature flags and configuration verified.
- Release notes published and the CHG ticket closed with the final outcome.

## 9. References

- [1] Nexora Coding Standards, NX-ENG-003.
- [2] Nexora API Documentation, NX-ENG-001.
- [3] Nexora Benefits Guide, NX-HR-003 (on-call allowance).
- [4] Google SRE Book: Managing Incidents; Release Engineering.

## 10. Revision History

**Table 4.** Document revision history.

| Version | Date        | Author                      | Summary of change                                            |
|---------|-------------|-----------------------------|--------------------------------------------------------------|
| 2.0     | 01 Oct 2025 | Nexora Platform Engineering | Introduced canary strategy and automatic rollback thresholds |
| 3.0     | 01 Jan 2026 | Nexora Platform Engineering | Added blackout periods and severity table                    |
| 3.1     | 01 Apr 2026 | Nexora Platform Engineering | Clarified emergency hotfix approval and on-call reachability |